Synthetic Acoustic Environments and Anechoic Siren Data: A Comprehensive Analysis of Acquisition, Procedural Generation, and Physics-Based Simulation for Autonomous Systems

1. Introduction: The Criticality of Anechoic Baselines in Machine Listening

The rapid and accelerating advancement of machine listening technologies, particularly within the specialized domains of autonomous driving (AD) and smart city infrastructure, has precipitated a critical and growing demand for high-fidelity acoustic training data. As vehicles transition from Level 2 driver assistance to Level 4 and 5 full automation, the sensory redundancy required to ensure safety in complex urban environments becomes paramount. While computer vision and LiDAR systems provide high-resolution spatial mapping, they are fundamentally limited by line-of-sight constraints. Acoustic perception systems, conversely, possess the unique capability to detect and localize salient events—such as approaching emergency vehicles—around corners, through occlusions, and at distances exceeding visual horizons. Among the myriad sound events necessary for the safe operation of autonomous vehicles, the detection and precise localization of emergency vehicle sirens—police cruisers, ambulances, and fire apparatus—stand as non-negotiable safety requirements.

However, the efficacy of Sound Event Detection (SED) and Direction of Arrival (DoA) estimation models is fundamentally constrained by the acoustic quality, diversity, and environmental neutrality of the training data. A significant bottleneck in the development of robust acoustic perception systems is the scarcity of "anechoic" or "dry" recordings of emergency sirens in open-access repositories. "Anechoic" refers to audio captured in an environment free from reflections, reverberation, and background noise, essentially isolating the source signal in its purest form. Such data is essential because it serves as the foundational "ground truth" upon which various acoustic environments can be artificially superimposed during training.

Most existing open-access datasets, such as UrbanSound8K, AudioSet, and ESC-50, predominantly feature "wet" recordings. These are audio signals that have already been convolved with the impulse response of a specific physical environment—such as a narrow city street, a wide highway, or a tunnel—and are often contaminated by uncontrollable background noise like wind, traffic hum, or overlapping speech. For robust machine learning

applications, particularly those utilizing advanced data augmentation strategies like Scaper or the physics-based KineScaper framework, the ideal input is a dry source signal. This allows researchers to artificially impose specific, controlled environmental characteristics—such as inverse-square law distance attenuation, Doppler shifts caused by relative motion, and frequency-dependent atmospheric absorption—without the interference of pre-existing acoustic artifacts that would otherwise compound and degrade the training signal.[1]

This comprehensive report investigates the current landscape of open-access anechoic siren datasets, exposing the limitations of field-recorded data for high-fidelity simulation. It posits that in the absence of extensive libraries of anechoic field recordings, the domain is shifting inevitably toward high-fidelity procedural synthesis. By leveraging the strictly regulated frequency and modulation parameters defined by standards such as SAE J1849 in the United States and similar regulations in Europe, researchers can algorithmically generate anechoic siren waveforms that are indistinguishable from recorded ground truth for the purposes of algorithmic training. Furthermore, this report will dissect the mathematical models governing these syntheses, evaluate the few existing repositories containing isolated siren events, and explore the physics-based simulation frameworks—such as PyRoadAcoustics—that transform these dry signals into realistic, dynamic, and annotated urban soundscapes suitable for training the next generation of autonomous perception systems.

2. The Physics and Standardization of Emergency Signaling

To understand the stringent requirements for anechoic siren data, one must first analyze the signal characteristics defined by regulatory bodies and the physics of sound generation. Emergency sirens are not random acoustic events; they are generated by precise electromechanical or electronic oscillators adhering to strict frequency sweep, modulation, and amplitude protocols. These protocols ensure that synthetic generation is a viable—and often superior—alternative to anechoic recording, as the "ground truth" is mathematically deterministic and legally standardized.

2.1 Spectral Characteristics and Modulation Patterns

The psychoacoustic efficacy of a siren is determined by its ability to penetrate ambient background noise (masking) and convey an immediate sense of urgency to the human auditory system. Acoustically, this is achieved through frequency modulation (FM) of a carrier wave. In mechanical sirens, this carrier is typically a complex wave approaching a sawtooth or distorted square wave, generated by a rotor chopping an air stream. In modern electronic systems, the carrier is often a synthesized square or sine wave, which is then amplified.[4]

The fundamental frequency (f_0) of a standard siren generally oscillates between a low bound (typically 500–700 Hz) and a high bound (1500–2000 Hz). This range is selected to maximize human auditory sensitivity, which peaks between 2 kHz and 5 kHz, while avoiding the

rapid atmospheric absorption that severely attenuates higher frequencies over long distances.[5]

The distinguishing feature between different siren modes—primarily "Wail," "Yelp," and "Phaser"—is the rate of this modulation. The "Wail" is characterized by a slow frequency sweep, typically cycling 10 to 30 times per minute. This slow rise and fall is designed to maximize long-range propagation and allow listeners to perceive the Doppler shift, aiding in localization and determining the direction of travel. The "Yelp" utilizes a much more rapid modulation rate, typically 150 to 250 cycles per minute. This rapid oscillation creates a sense of high urgency and facilitates detection in short-range, high-traffic density scenarios where rapid fluctuations help the signal cut through constant-frequency masking noise, such as engine hum or tire roar.[7]

Table 1: Standard Acoustical Parameters for Common Siren Modes

Siren Mode	Modulation Rate (Cycles/Min)	Frequency Range (Hz)	Waveform Type	Psychoacoustic Function
Wail	10 – 30	500 – 1600	Square / Trap.	Long-range warning; distinct rise/fall allows Doppler estimation.
Yelp	150 – 250	500 – 1600	Square / Trap.	High urgency; intersection clearing; cuts through traffic noise.
Hi-Lo	60 – 90	450 / 600 (Discrete)	Square	European standard (Two-Tone); distinct interval (often minor third).
Phaser/Piercer	600 – 900	1500 – 2500	Square	Extreme urgency; highly

				dispersive; rapid modulation.
Rumbler	Matches Primary	125 – 400	Sine / Square	Visceral vibration; penetrates solid structures (vehicle bodies).

Data synthesized from SAE J1849 standards, California Title 13 regulations, and manufacturer specifications.[9]

2.2 Regulatory Standards: SAE J1849 and Title 13

In the United States, the Society of Automotive Engineers (SAE) establishes the J1849 standard, which specifies the laboratory performance requirements for sirens. Crucially for this analysis, these standards require measurements to be taken in anechoic chambers to ensure the pure output of the device is characterized without environmental reflections. For a Class A siren classification under California Title 13, the device must produce a minimum sound pressure level (SPL) of 120 dB on the acoustic axis at a distance of 3 meters.[8]

The standard mandates that the fundamental frequency of the siren must sweep within specific bounds. For a "Wail," the frequency must cycle automatically between a low and high point at a rate of 10 to 30 oscillations per minute. For a "Yelp," the rate increases to 150–250 oscillations per minute.[8] The sound level must not drop more than 10 dB below the peak values during the lowest portion of the cycle. These precise definitions are critical for researchers generating synthetic datasets; a Python script generating a sine sweep from 600 Hz to 1500 Hz with a 4-second period is not merely an approximation but a regulatory-compliant reconstruction of a Wail signal that meets the legal definition of an emergency signal in California.

European standards, such as CEN 1789 for ambulances, often mandate the "Hi-Lo" or "Two-Tone" pattern. Unlike the continuous frequency sweep of the American Wail/Yelp, the European siren alternates between two discrete frequencies. Research indicates that the British "Wail" siren operates at a higher pitch (1,000–3,000 Hz) compared to the continental European Hi-Lo (450–600 Hz), making it more intrusive but potentially more susceptible to attenuation over distance.[12] Understanding these regional differences is vital for creating global datasets; a model trained solely on US-style synthetic wails may fail to detect the

discrete interval of a German police siren.

2.3 The "Rumbler" and Low-Frequency Propagation

A significant development in modern emergency signaling is the introduction of the "Rumbler" or "Howler" system. These secondary sirens emit low-frequency tones (typically 125–400 Hz) that operate in tandem with the primary high-frequency siren. The physical principle behind this is the longer wavelength of low-frequency sound, which is less susceptible to diffraction by vehicle bodies and absorption by glass and modern sound-insulation materials.[9]

In the context of anechoic data generation, simulating a Rumbler involves generating a sub-harmonic waveform that tracks the modulation of the primary siren but at a significantly lower octave, often two octaves down. This represents a distinct class of sound event that is virtually absent from older datasets like UrbanSound8K but is increasingly relevant for autonomous vehicle perception systems that rely on diverse acoustic signatures to detect vehicles in blind spots. The Rumbler's ability to vibrate the physical structure of nearby vehicles provides a tactile cue that is often modeled in multi-modal datasets but requires specific acoustic synthesis to replicate in audio-only training.

3. Availability and Limitations of Open Access Datasets

The user query specifically targets "anechoic siren sound dataset open access." A comprehensive review of the research material reveals a stark dichotomy: while there is an abundance of *urban soundscape* data containing sirens, there is a profound scarcity of *anechoic* or *dry* siren recordings. This necessitates a detailed examination of what is available and the limitations inherent in using field recordings for precise acoustic simulation.

3.1 The Scarcity of Anechoic Data

Anechoic recordings are captured in specialized chambers lined with absorptive wedges that eliminate all reflected sound, effectively simulating a free-field environment. Acquiring such data for sirens is logistically difficult; it requires driving an emergency vehicle into a large anechoic chamber or mounting a high-output siren driver in a smaller chamber. Consequently, most "clean" siren samples found in open datasets are actually near-field recordings taken outdoors, which still contain ground reflections and some environmental noise, or they are synthesizer outputs.

The **EARS (Expressive Anechoic Recordings of Speech)** dataset [13] and the **Anechoic Interferer Dataset (AID)** [14] demonstrate the existence of high-quality anechoic resources for speech and domestic sounds. AID, for example, provides dry recordings of various domestic noise sources to serve as interferers. However, no comparable large-scale repository exists specifically for emergency vehicle sirens. The **sireNNet** dataset [15] is a prominent resource,

containing categorized samples of ambulance, police, and fire truck sirens. However, the methodology for sireNNet involved scraping audio from the internet (YouTube, etc.), meaning the samples are inherently "wet"—they include the acoustic imprint of the recording location (Doppler shifts from motion, city reverb, traffic noise). While useful for classification tasks, these recordings are suboptimal for physics-based augmentation because the acoustic environment is "baked in," making it impossible to accurately simulate a new acoustic environment without first performing complex and imperfect de-reverberation.

3.2 Key Urban Sound Datasets Containing Sirens

Despite the lack of pure anechoic libraries, several datasets serve as de facto standards for siren detection research. These datasets often provide the "background" or "noise" component for synthetic mixing, or contain siren instances that can be extracted and de-noised.

Table 2: Comparative Analysis of Major Open Access Siren-Inclusive Datasets

Dataset	Nature of Audio	Siren Representation	Suitability for Anechoic Simulation	Access/License
UrbanSound8K [2]	Field Recordings	~1,000 labeled excerpts (class 8). High variability.	**Low.** Contains significant reverb and background noise.	CC BY-NC 4.0
AudioSet [3]	YouTube Excerpts	Extensive (thousands of clips). Ontology includes "Siren", "Police car", etc.	**Low.** Data is heavily compressed and noisy. Useful for robust validation.	CC BY 4.0 (Features)
sireNNet [15]	Web Scraped	Curated specifically for sirens. High class diversity (Fire, Police, Ambulance).	**Medium-Low.** Sourced from video; mostly wet audio with Doppler shifts present.	CC BY 4.0

MAVD [16]	Urban Field Rec.	Traffic focused. Includes "Siren" as a sound event.	**Low.** Focus is on realistic traffic noise and visual alignment.	CC BY 4.0
FSD50K [18]	Freesound (Crowd)	Variable. Contains high-quality user uploads, some near-anechoic .	**Medium.** requires filtering for "dry" tags.	CC BY 4.0
TUT Rare Sound 2017 [19]	Synthetic Mixtures	Contains "target" isolated events (including sirens) for mixture creation.	**High.** Provides source files intended for mixing, often cleaner than field recs.	Research/Edu
Kaggle Siren Sounds [20]	Web Scraped	600 files (Ambulance, Fire, Traffic). Short 3s clips.	**Low.** Manual extraction from internet sources.	Database Dependent

3.3 The "Wet" Data Problem in Simulation

The primary limitation of datasets like UrbanSound8K for simulation purposes is the presence of environmental convolution. If a researcher wishes to simulate an ambulance approaching from 100 meters away in a virtual city, using a recording from UrbanSound8K is problematic because that recording might already contain the reverberation of a narrow street in New York City. Superimposing a second set of reverberations (from the simulation) onto the first (from the recording) results in an unnatural acoustic artifact often referred to as "double-reverberation," which can degrade the performance of models trained on such data when deployed in the real world.

Furthermore, most field recordings capture the siren passing by, meaning the **Doppler shift** is already present in the audio. The "KineScaper" framework [1] highlights this as the "SNR Problem" in dynamic mixing. If the source audio already fades in and out (due to the recording vehicle moving), the physics engine cannot accurately apply its own distance attenuation laws

($p(r) \propto 1/r$). For accurate simulation, the source file must be steady-state: constant amplitude and constant pitch (or fixed modulation cycle) without any pre-existing Doppler shift. Using a recording that already contains a Doppler shift as a source for a new simulation would result in physically impossible frequency shifts.

4. Procedural Synthesis: Generating the Anechoic Baseline

Given the scarcity of recorded anechoic data, the research community increasingly relies on **procedural synthesis**. This involves using digital signal processing (DSP) techniques to generate siren tones from mathematical first principles. This approach yields data that is definitionally anechoic, perfectly clean, and infinitely variable within the bounds of regulatory standards. This effectively solves the "wet data" problem by creating a pristine source signal that can be manipulated at will.

4.1 Mathematical Modeling of Siren Tones

The fundamental building block of a siren synthesizer is the voltage-controlled oscillator (VCO). In a digital domain, this is represented by a waveform generator whose frequency is modulated over time.

For a standard **Wail** siren, the frequency $f(t)$ varies continuously between a minimum (f_{min}) and maximum (f_{max}) frequency over a period T. This can be modeled using a sinusoidal modulation of the frequency, although in many electronic sirens, the sweep is linear (triangle wave modulation) or exponential. A basic Python implementation using numpy and scipy can generate these waveforms with high precision.[4] The core equation for a frequency-modulated signal is:

$$y(t) = A \cdot \sin(2\pi \cdot \phi(t))$$
Where $\phi(t)$ is the instantaneous phase, which is the integral of the instantaneous frequency $f(t)$:

$$\phi(t) = \int_{0}^{t} f(\tau) d\tau$$
If the frequency sweeps linearly from f_{start} to f_{end} over time T, the instantaneous frequency is:

$$f(t) = f_{start} + \frac{f_{end} - f_{start}}{T} t$$
This allows for the generation of a **linear chirp**, which forms the rising or falling edge of a siren. By concatenating a rising chirp and a falling chirp, one constructs a complete Wail or Yelp cycle. The Python library scipy.signal.chirp is frequently used for this purpose.[22] The

modulation index and carrier characteristics can be tweaked to simulate different siren models (e.g., Federal Signal vs. Whelen), providing a diversity of training data that would be expensive to collect physically.

4.2 Python-Based Procedural Generation

Several open-source repositories and scripts facilitate this generation, serving as effective "synthetic datasets" where the code itself is the generator of the data. This "data as code" approach ensures reproducibility and allows researchers to generate infinite variations.

1. **Basic FM Synthesis:** Scripts utilizing math.sin or numpy.sin with a time-varying frequency parameter can create convincing Wail and Yelp tones. The modulation index and carrier frequency are adjusted to match SAE J1849 standards (e.g., sweeping 500Hz-1500Hz).[4]
2. **Scipy Signal Processing:** The scipy.signal module provides robust tools for waveform generation. Specifically, creating a sawtooth or square wave carrier (which mimics the harsh timbre of mechanical sirens) and modulating its frequency is a common technique.[23] This allows for the inclusion of harmonics that are present in real mechanical sirens but absent in simple sine wave synthesis.
3. **Dedicated Libraries:**
    - **Pyo:** A digital signal processing module for Python that allows for complex, real-time audio synthesis. It can simulate envelopes, frequency modulation, and even the "dirt" or noise associated with mechanical rotors.[24]
    - **PySineWave:** A lighter weight library for generating sine waves with smooth pitch transitions, useful for creating the glissando effects of wailing sirens.[25]
    - **SirenGenerator (GitHub):** Repositories such as arda-guler/siren_generator provide ready-made Python classes to output specific alarm tones, effectively acting as a procedural dataset generator.[26]

4.3 Advantages of Synthetic Anechoic Data

Using algorithmically generated sirens offers distinct advantages over recorded data for machine learning training:

- **Perfect Labels:** The exact onset, offset, frequency, and modulation rate are known ground truths, eliminating annotation errors and ambiguity.
- **Parameter Sweeping:** A researcher can generate thousands of variations of a "Wail" by slightly perturbing the f_{min}, f_{max}, and sweep rate parameters. This improves model robustness to the variances found in different hardware manufacturers.
- **Aliasing and Artifact Control:** Synthetic generation allows for precise control over sampling rates (e.g., 44.1kHz vs 48kHz) and bit depth, ensuring the training data is of the highest possible fidelity before degradation is applied during augmentation.
- **Separation of Concerns:** By starting with a synthetic signal, any noise or reverberation added later is fully controllable and labeled, allowing for the training of de-noising and

de-reverberation models alongside detection models.

5. Physics-Based Propagation and Environmental Simulation

Once a dry/anechoic siren signal is obtained (either via recording or synthesis), the next step in creating a training dataset for autonomous vehicles is **propagation simulation**. This transforms the sterile anechoic sound into a realistic representation of what a vehicle's microphone array would capture in a dynamic urban environment. This step bridges the gap between the "simulated" and "real" domains (Sim2Real).

5.1 The KineScaper and PyRoadAcoustics Frameworks

The document "KineScaper: A Physics-Based Dynamic Soundscape Generator" [1] provides a theoretical blueprint for this process. It argues for a "Physics-First" approach to augmentation, distinct from simple additive mixing. Unlike simple mixing where a sound is just added to a background, physics-based simulation models the journey of the sound wave through the environment.

Key Components of Simulation:

1. **Inverse Square Law (Amplitude):** The sound pressure level (SPL) decreases as the distance r from the source increases, proportional to $1/r$. This is the primary cue for distance estimation.
2. **Atmospheric Absorption:** High frequencies are attenuated more rapidly by the atmosphere than low frequencies. This acts as a distance-dependent low-pass filter (ISO 9613-1). This is critical for sirens, as the "sharpness" of the sound diminishes over distance, a cue humans use to estimate range.[1] The simulation must account for temperature and humidity, which affect absorption rates.
3. **Doppler Effect:** As the emergency vehicle moves relative to the observer, the received frequency shifts. PyRoadAcoustics simulates this using Variable Delay Lines (VDL). The delay $D[n]$ varies over time based on the relative velocity vectors of the source and receiver.[6] This captures the characteristic pitch drop of a passing vehicle.

PyRoadAcoustics [28] is identified as a primary tool for this simulation. It is an open-source Python library specifically designed to simulate the sound propagation of vehicles on a road. It models:

- **Direct Sound:** The straight path from siren to microphone.
- **Ground Reflection:** The "acoustic shadow" and interference patterns caused by sound bouncing off the asphalt. This is modeled using complex reflection coefficients that depend on the road surface material.
- **Source Directivity:** Sirens are not isotropic; they project sound forward. PyRoadAcoustics allows for directivity patterns to be applied, simulating the difference

between a siren facing the listener versus one driving away.[30]

5.2 The "SNR Problem" and Mixing Strategies

A sophisticated insight from the KineScaper proposal [1] is the challenge of defining Signal-to-Noise Ratio (SNR) for moving sources. In static detection tasks, SNR is constant. For a passing ambulance, SNR varies continuously from $-\infty$ dB (inaudible distance) to $+30$ dB (close pass) and back. A single scalar SNR value is insufficient to describe a dynamic event.

To address this, modern simulation pipelines utilize a **"Generate First, Filter Later"** strategy. Instead of forcing a specific input SNR, the simulation places the source on a trajectory and calculates the resulting levels based on physics. Post-generation analysis then frames the audio and calculates local SNR for each frame, providing granular labels (e.g., snr_min, snr_max) that are highly valuable for training regression models that predict proximity.[1] This output-based analysis ensures that the labels reflect the physical reality of the simulation, rather than an arbitrary input parameter.

5.3 Acoustic Scene Composition

To create a complete dataset, the propagated anechoic siren is mixed with "background" textures. The **Sound of Traffic** [31] and **MAVD** [16] datasets serve as excellent sources for these backgrounds. They provide long-duration recordings of traffic noise, urban hum, and environmental sounds. These datasets, while not anechoic themselves, offer high-fidelity recordings of the *environment* without the target *event* (the siren). By mixing the dynamic, physics-simulated siren tracks into these real-world backgrounds, researchers create **Augmented Urban Soundscapes**.[33] This hybrid approach—synthetic foregrounds, real backgrounds—is currently the gold standard for training robust SED models for automotive applications, offering the realism of field recordings with the control of synthesis.

6. Machine Learning Applications and Model Architectures

The ultimate consumer of these datasets is the machine learning model designed to detect and classify sirens. The shift from real-world data collection to synthetic/hybrid generation has profound implications for model architecture and performance. The availability of clean, parameterized data allows for more specialized model training strategies.

6.1 Sound Event Detection (SED) in Automotive Environments

The primary task is identifying the presence of a siren within the ego-vehicle's cabin. This is acoustically complex due to the "sound shielding" effect of the car chassis and the presence of competing noise (engine, HVAC, infotainment). Models like **SirenNet** [15] and **E2PANNs** [34]

have been developed specifically for this binary or multi-class classification problem. The use of synthetic data allows these models to be trained on edge cases that are dangerous or difficult to record, such as a siren approaching a high-speed intersection.

Recent research proposes small-footprint solutions suitable for embedded devices (Edge Computing). For instance, a 2025 study proposes a model with only 1,260 parameters using depth-separable convolutional layers and Gated Recurrent Units (GRUs), achieving high accuracy on synthetically generated datasets.[35] This validates the efficacy of synthetic training data: models trained on physics-based simulations generalize well to real-world scenarios because the simulations accurately capture the temporal-spectral evolution of the Doppler-shifted siren, which is the most robust feature for detection.

6.2 Data Efficiency and Transfer Learning

The challenge of "Data-Efficient Low-Complexity Acoustic Scene Classification" (DCASE Task 1) [36] highlights the trend toward training on smaller, high-quality datasets. Anechoic synthesis allows for **Curriculum Learning**, where models are first trained on high-SNR, clean synthetic sirens to learn the fundamental features (harmonic stacks, sweep rates), and then progressively exposed to lower-SNR, highly reverberant mixtures. This mimics the human learning process and has been shown to improve convergence and robustness in deep learning models.

6.3 Voice Activity Detection (VAD) and Disambiguation

An emerging theme in the literature is the need to disambiguate sirens from other harmonic sources, particularly music and speech. Since sirens often sit in the vocal frequency range (500Hz-2kHz), simple energy detection fails. Advanced datasets now include specific "distractor" classes. The synthesis pipelines described in Section 5 allow for the controlled injection of speech (e.g., from the EARS dataset) to train models that can distinguish between a singing voice on the radio and an external wail siren.[37] This disambiguation capability is critical to preventing false positives in real-world deployment.

7. Regulatory and Legal Context

The generation and use of siren datasets must be understood within the framework of legal standards, which dictate not only the acoustic properties of the sirens but also the testing methodologies. These regulations essentially define the "ground truth" for synthetic generation.

7.1 US Standards: SAE and Title 13

As previously noted, **SAE J1849** is the governing standard in the US. It requires sirens to be tested in anechoic chambers to verify their output levels (typically >118 dB at 3m). California's **Title 13** imposes additional constraints on the sweep rates of Wail and Yelp tones. These

regulations effectively provide the "blueprint" for procedural synthesis. A valid synthetic dataset must adhere to these sweep rates (e.g., Wail at 10-30 cpm) to be considered a realistic representation of US emergency vehicles.[9] Failing to adhere to these parameters results in a dataset that trains the model on "invalid" features.

7.2 European Standards: CEN 1789

In Europe, **CEN 1789** governs ambulance specifications. The acoustic requirements often favor the "Two-Tone" (Hi-Lo) signal over the continuous sweep used in the US. This fundamental difference means that a model trained on US data (Wail/Yelp) will likely fail to detect European sirens, and vice versa. Therefore, global datasets must explicitly label the siren type and region. The **sireNNet** dataset attempts to address this by including diverse classes, but the lack of geographic metadata in web-scraped data remains a limitation.[38] Procedural generation allows researchers to specifically target regional standards by simply changing the modulation parameters in the synthesis script.

7.3 Compliance in Simulation

For a simulation to be valid for safety certification (e.g., testing an AV's ability to yield to emergency vehicles), the virtual siren must meet the legal standards of the operational domain. If an AV is deployed in California, the simulated siren used to validate its safety systems must comply with Title 13. This reinforces the value of procedural generation, where these parameters can be hard-coded and verified, ensuring legal compliance of the training data—something that cannot be guaranteed with "in the wild" recordings of unknown origin.

8. Conclusion

The search for "anechoic siren sound datasets" reveals a significant gap in open-access repositories: while field recordings are plentiful, pure anechoic data is rare. This scarcity has driven the field of computational acoustics toward a hybrid paradigm of **Procedural Synthesis** and **Physics-Based Simulation**. This shift is not merely a workaround but an evolution toward higher-fidelity, more controllable, and legally compliant training data.

By utilizing Python-based DSP libraries to generate waveforms that strictly adhere to SAE J1849 and CEN 1789 standards, researchers can create "perfect" anechoic source material. Tools like **PyRoadAcoustics** and **KineScaper** then serve as the bridge, applying mathematically rigorous propagation models—including Doppler shift, atmospheric absorption, and ground reflection—to place these synthetic sirens into realistic virtual environments.

This approach offers distinct advantages over field recordings: it solves the ground truth labeling problem, allows for infinite variability in training data (Data Augmentation), and ensures regulatory compliance. For the development of robust acoustic perception systems in autonomous vehicles, the future lies not in finding more recordings of sirens, but in generating

better simulations of them.

Recommendations for Researchers

1. **Adopt Procedural Generation:** Do not rely solely on web-scraped siren audio, which is often contaminated with reverb and noise. Use Python scripts to generate Wail/Yelp/Hi-Lo tones based on standard frequencies (500Hz-1600Hz) and modulation rates.
2. **Utilize Propagation Simulators:** Leverage libraries like pyroadacoustics to artificially induce Doppler effects and distance attenuation. This creates a "dynamic" dataset from static source files that accurately reflects the physics of a moving vehicle.
3. **Hybridize Data:** Mix these synthetic events into high-quality, long-duration urban backgrounds (e.g., from the **Sound of Traffic** or **MAVD** datasets) to create realistic SNR scenarios.
4. **Validate with Standards:** Ensure generated waveforms comply with the specific legal standards (Title 13 vs. CEN 1789) relevant to the deployment region of the target application to ensure the model learns valid features.

9. Appendix: Technical Reference for Procedural Generation

9.1 Mathematical Definition of Siren Tones

Wail Tone:

- **Waveform:** Square or distorted sine.
- **Frequency Range:** $f_{low} \approx 600 \text{ Hz}$, $f_{high} \approx 1600 \text{ Hz}$.
- **Period (T):** ≈ 4 seconds (15 cycles/min).
- **Function:** $f(t) = f_{low} + (f_{high} - f_{low}) \cdot \frac{1}{2}(1 - \cos(\frac{2\pi t}{T}))$ (for sinusoidal sweep).

Yelp Tone:

- **Waveform:** Square or distorted sine.
- **Frequency Range:** $f_{low} \approx 600 \text{ Hz}$, $f_{high} \approx 1600 \text{ Hz}$.
- **Period (T):** ≈ 0.3 seconds (200 cycles/min).
- **Function:** Rapid linear or exponential sweep.

9.2 Python Implementation Strategy

To implement a procedural generator [4]:

1. **Time Array:** Generate a time vector t using numpy.linspace with sampling rate $f_s = 44100$ Hz.
2. **Modulator:** Create the frequency envelope array f_t using scipy.signal.sawtooth or

numpy.sin to oscillate between f_{low} and f_{high} at the desired rate.
3. **Phase Integration:** Compute phase phi = 2 * np.pi * np.cumsum(f_t) / fs.
4. **Carrier Generation:** Generate the audio signal y = amplitude * signal.square(phi) (or np.sin).
5. **Output:** Save as WAV using scipy.io.wavfile.write.

This procedural pipeline ensures the creation of truly anechoic, noise-free siren data suitable for high-fidelity acoustic simulation and machine learning training.

9.3 Table of Reviewed Open Access Datasets

Dataset Name	Content Type	Key Features	Access Link
UrbanSound8K	Field Recordings	10 classes, includes sirens. Reverb present.	innovatiana.com [2]
sireNNet	Web Scraped	Emergency vehicles specific. Categorized.	(https://data.mendeley.com/datasets/j4ydzzv4kb) [15]
AudioSet	YouTube Audio	Massive scale, includes "Siren", "Emergency vehicle".	research.google.com [39]
MAVD	Audio-Visual	Traffic focus, vehicle components.	Zenodo [40]
IDMT-Traffic	Field Recordings	Vehicle pass-bys, stereo, high quality.	(https://www.idmt.fraunhofer.de/en/publications/datasets/traffic.html) [41]
Sound of Traffic	Field Recordings	415 hours, multi-channel, diverse traffic.	Zenodo [42]
TUT Rare Sound	Synthetic Mixtures	Isolated target	Zenodo [43]

		events + backgrounds.	
AID	Anechoic Recordings	Domestic sounds (interferers), pure anechoic.	AudioLabs Erlangen [14]

Note: While AID provides anechoic data, it focuses on domestic sounds. For sirens, procedural generation remains the primary method for obtaining anechoic data.

Bibliografia

1. Audio Mixing Project Brainstorm.pdf
2. UrbanSound8k Dataset: Recognizing urban sounds | Innovatiana, accesso eseguito il giorno gennaio 14, 2026, https://www.innovatiana.com/en/datasets/urbansound8k
3. The AudioSet dataset - Google Research, accesso eseguito il giorno gennaio 14, 2026, https://research.google.com/audioset/dataset/index.html
4. Creating police siren sounds with frequency modulation - Applied Mathematics Consulting, accesso eseguito il giorno gennaio 14, 2026, https://www.johndcook.com/blog/2016/03/10/creating-police-siren-sounds-with-frequency-modulation/
5. Loud and Clear: The Science and Strategy Behind Emergency Vehicle Sirens, accesso eseguito il giorno gennaio 14, 2026, https://www.blueprintfleet.com/drawingboard/loud-and-clear-the-science-and-strategy-behind-emergency-vehicle-sirens
6. pyroadacoustics: a road acoustics simulator based on variable length delay lines, accesso eseguito il giorno gennaio 14, 2026, https://www.dafx.de/paper-archive/2022/papers/DAFx20in22_paper_9.pdf
7. Different Siren Sounds Used by First Responders - LED Equipped, accesso eseguito il giorno gennaio 14, 2026, https://www.ledequipped.com/blogs/news/different-siren-sounds-used-by-first-responders
8. Cal. Code Regs. Tit. 13, § 1028 - Performance Requirements | State Regulations | US Law, accesso eseguito il giorno gennaio 14, 2026, https://www.law.cornell.edu/regulations/california/13-CCR-1028
9. Sound Solutions: Choosing the Right Siren for Your Public Safety Fleet - Police, accesso eseguito il giorno gennaio 14, 2026, https://www.government-fleet.com/10234309/sound-solutions-choosing-the-right-siren-for-your-public-safety-fleet
10. Federal Signal, 650007 Siren, 100W, 12VDC - Wail and Yelp Tone - LEHR Auto, accesso eseguito il giorno gennaio 14, 2026, https://lehrauto.com/sirens-controllers/sirens/federal-signal-650007-siren-100w-12vdc-wail-and-ye-650007

11. FR100 Series Siren - SoundOff Signal, accesso eseguito il giorno gennaio 14, 2026, https://www.soundoffsignal.com/s/product/fr100-series-siren/01t5f0000044tAsAAI
12. The sound of sirens - The New World - The New European, accesso eseguito il giorno gennaio 14, 2026, https://www.thenewworld.co.uk/paola-totaro-the-sound-of-sirens/
13. EARS: An Anechoic Fullband Speech Dataset Benchmarked for Speech Enhancement and Dereverberation - GitHub Pages, accesso eseguito il giorno gennaio 14, 2026, https://sp-uhh.github.io/ears_dataset/
14. AID: Open-Source Anechoic Interferer Dataset - International Audio Laboratories Erlangen, accesso eseguito il giorno gennaio 14, 2026, https://www.audiolabs-erlangen.de/resources/2022-IWAENC-Anechoic-Noise-Dataset
15. sireNNet-Emergency Vehicle Siren Classification Dataset For Urban Applications - Mendeley Data, accesso eseguito il giorno gennaio 14, 2026, https://data.mendeley.com/datasets/j4ydzzv4kb
16. (PDF) MAVD: A Dataset for Sound Event Detection in Urban Environments - ResearchGate, accesso eseguito il giorno gennaio 14, 2026, https://www.researchgate.net/publication/336903988_MAVD_A_Dataset_for_Sound_Event_Detection_in_Urban_Environments
17. mavd: a dataset for sound event detection in urban environments - DCASE, accesso eseguito il giorno gennaio 14, 2026, https://dcase.community/documents/workshop2019/proceedings/DCASE2019Workshop_Zinemanas_70.pdf
18. Datasets - Freesound Labs, accesso eseguito il giorno gennaio 14, 2026, https://labs.freesound.org/datasets/
19. Detection of rare sound events - DCASE, accesso eseguito il giorno gennaio 14, 2026, https://dcase.community/challenge2017/task-rare-sound-event-detection
20. Emergency Vehicle Siren Sounds - Kaggle, accesso eseguito il giorno gennaio 14, 2026, https://www.kaggle.com/datasets/vishnu0399/emergency-vehicle-siren-sounds
21. Generate Audio with Python - Zach Denton, accesso eseguito il giorno gennaio 14, 2026, https://zach.se/generate-audio-with-python/
22. Signal processing (scipy.signal) — SciPy v1.17.0 Manual, accesso eseguito il giorno gennaio 14, 2026, https://docs.scipy.org/doc/scipy/reference/signal.html
23. Sound Synthesis in Python. Additive synthesis from basics using... | by Noah Hradek, accesso eseguito il giorno gennaio 14, 2026, https://medium.com/@noahhradek/sound-synthesis-in-python-4e60614010da
24. How to Generate Sound Effects Programmatically Using Python and Pyo - DEV Community, accesso eseguito il giorno gennaio 14, 2026, https://dev.to/hexshift/how-to-generate-sound-effects-programmatically-using-python-and-pyo-2eid
25. pysinewave - PyPI, accesso eseguito il giorno gennaio 14, 2026, https://pypi.org/project/pysinewave/

26. siren-generator · GitHub Topics, accesso eseguito il giorno gennaio 14, 2026, https://github.com/topics/siren-generator
27. arda-guler/siren_generator: Python alarm tone/siren generator - GitHub, accesso eseguito il giorno gennaio 14, 2026, https://github.com/arda-guler/siren_generator
28. steDamiano/pyroadacoustics: Python package for road acoustics simulation based on variable length delay lines. - GitHub, accesso eseguito il giorno gennaio 14, 2026, https://github.com/steDamiano/pyroadacoustics
29. steDamiano/EV-sound-classification-dataset - GitHub, accesso eseguito il giorno gennaio 14, 2026, https://github.com/steDamiano/EV-sound-classification-dataset
30. Siren sound levels inside the ambulance. | Download Scientific Diagram - ResearchGate, accesso eseguito il giorno gennaio 14, 2026, https://www.researchgate.net/figure/Siren-sound-levels-inside-the-ambulance_fig4_6277580
31. Sound of Traffic: A Dataset for Acoustic Traffic Identification and Counting - ISCA Archive, accesso eseguito il giorno gennaio 14, 2026, https://www.isca-archive.org/interspeech_2024/ghaffarzadegan24_interspeech.pdf
32. Sound of Traffic: A Dataset for Acoustic Traffic Identification and Counting - ISCA Archive, accesso eseguito il giorno gennaio 14, 2026, https://www.isca-archive.org/interspeech_2024/ghaffarzadegan24_interspeech.html
33. (PDF) ARAUS: A Large-Scale Dataset and Baseline Models of Affective Responses to Augmented Urban Soundscapes - ResearchGate, accesso eseguito il giorno gennaio 14, 2026, https://www.researchgate.net/publication/368727300_ARAUS_A_Large-Scale_Dataset_and_Baseline_Models_of_Affective_Responses_to_Augmented_Urban_Soundscapes
34. [PDF] Large-scale audio dataset for emergency vehicle sirens and road noises, accesso eseguito il giorno gennaio 14, 2026, https://www.semanticscholar.org/paper/051cda432963a18f94867eb5b849cbd4eb712241
35. A Small-Footprint Deep-Learning Solution for Real-Time In-Car Emergency Vehicle Detection - Eusipco 2025, accesso eseguito il giorno gennaio 14, 2026, https://eusipco2025.org/wp-content/uploads/pdfs/0000156.pdf
36. Data-Efficient Low-Complexity Acoustic Scene Classification - DCASE, accesso eseguito il giorno gennaio 14, 2026, https://dcase.community/challenge2024/task-data-efficient-low-complexity-acoustic-scene-classification
37. EARS: An Anechoic Fullband Speech Dataset Benchmarked for Speech Enhancement and Dereverberation | Request PDF - ResearchGate, accesso eseguito il giorno gennaio 14, 2026, https://www.researchgate.net/publication/383653676_EARS_An_Anechoic_Fullband_Speech_Dataset_Benchmarked_for_Speech_Enhancement_and_Dereverberation

38. CEN 1789 - Wikipedia, accesso eseguito il giorno gennaio 14, 2026, https://en.wikipedia.org/wiki/CEN_1789
39. Emergency vehicle - AudioSet, accesso eseguito il giorno gennaio 14, 2026, https://research.google.com/audioset/ontology/emergency_vehicle_1.html
40. MAVD-traffic dataset - Zenodo, accesso eseguito il giorno gennaio 14, 2026, https://zenodo.org/records/3338727
41. IDMT-TRAFFIC, accesso eseguito il giorno gennaio 14, 2026, https://www.idmt.fraunhofer.de/en/publications/datasets/traffic.html
42. DCASE 2024 Challenge Task 10 Development Dataset: Acoustic-based Traffic Monitoring, accesso eseguito il giorno gennaio 14, 2026, https://zenodo.org/records/10700792
43. TUT Rare sound events, Development dataset - Zenodo, accesso eseguito il giorno gennaio 14, 2026, https://zenodo.org/records/401395